COSH: Closed-Form Onset Search via Hyperbolic Response

Methodology for pre-flight critical-moment identification of a multirotor from ground-contact tip-over excitation

This document specifies the complete identification methodology, from raw signals to the take-off feed-forward moment. Every stage mirrors the implementation in `critical_value_getter_piecewise.py`; the companion analyses (model-fidelity bounds, excitation design, ground-effect bracket) are documented in `fidelity_section.tex` and are only referenced here.

1 Problem setting and notation

A multirotor rests on its landing gear. Ramping the commanded roll (pitch) moment tilts the vehicle until it begins to rotate about the landing-gear contact line — the *pivot*. The applied moment at that *rotation onset* is the critical moment M_{crit}; comparing the two tip directions yields the CoM offset and the feed-forward moment M_{ff} that compensates take-off. No rig, no motion capture and no stable in-flight data are used.

Symbol	Meaning	Value	Provenance
W, z_{CoM}	weight, CoM height	31.59 N, 0.261 m	measured / CAD
	(case 1, unloaded)	30.08 N	measured
$l_{p,\phi}$, $l_{p,\theta}$	pivot offsets (roll, pitch)	0.140, 0.110 m	mocap arc fit
J_P	inertia about the pivot line	—	never used numerically
C_T	thrust map, $T_i = C_T\Omega_i^2$	1.3175×10^{-7}	bench calibration
M_{thr}	window threshold	0.01 N m	noise floor
M_{cap}	allocator caps (x, y)	2.37, 2.74 N m	allocation feasibility
$\varepsilon_{\max}$	ramp-slope gate	3%	sensitivity chain (Sec. 5)
M_{floor}	slope-gate floors (x, y)	0.7, 0.4 N m	$\lvert M_{\text{crit}}\rvert$ lower bound, $\pm$20 mm box
$N_{\min}$	window-sample gate	38	30 pre + 8 post
$e_{\text{lin},\max}$	linearity-RMSE gate	30 mN m	10× noise floor
N_{seg}	sweep margin (each end)	8	degenerate-candidate guard

Table 1: Design constants. (C_2, K) are *calibrated*, not set (Sec. 7).

2 Pipeline overview

The complete pipeline is five stages (Fig. 1); the three algorithms nest as Algorithm 3 ⊃ Algorithm 2 ⊃ Algorithm 1, and Algorithm 1 is the only stage that touches the angular rate.

1. **Signals and window** (Sec. 4) — per run: reconstruct $M(t)$ from the measured rotor speeds and take $\omega(t)$ from odometry; open the excitation window at $|M| > M_{\text{thr}}$ (backward from the peak), close it at the moment peak, truncated at the allocator cap.
2. **Run gates** (Sec. 5) — discard runs failing the slope-error (3%), sample-count (38) or linearity-RMSE (30 mN m) gates; moment trace only, so no circularity.
3. **Calibration, once per configuration and axis** (Sec. 7) — a per-run free fit (PNLS) runs first and its per-dataset medians locate (C_2, K); Algorithm 3 then refines that centre on a grid, and each candidate is scored by Algorithm 2, which runs Algorithm 1 on every gated run and sums the two directional CVs of M_{crit} across ramp rates.

4. **Identification, per run** (Sec. 6) — Algorithm 1 with the calibrated (C_2^*, K^*) pinned, at full sweep resolution: amplitude fixed by $C_1 = K\dot{M}$, baseline by the pre-segment median, the onset index swept exhaustively; $M_{\text{crit}} = M(t_{j^*})$.

5. **Aggregation** (Sec. 8) — directional means $\bar{M}_\pm$, the pivot-free feed-forward moment $M_{\text{ff}} = \frac{1}{2}(\bar{M}_+ + \bar{M}_-)$, and the CoM offset for validation against the load-cell truth (Sec. 9).

3 Contact-phase dynamics and the closed form

About the active pivot line, with the small-angle attitude dependence retained ($\cos\varphi \approx 1$, $\sin\varphi \approx \varphi$), the roll/pitch dynamics read

$$J_P\,\dot{\omega} = M(t) + f\,l_p - W\,(l_p + \lambda_{\text{off}}) + W z_{\text{CoM}}\,\varphi, \tag{1}$$

where M is the commanded body moment, f the collective thrust and λ_{off} the CoM offset along the excited axis. Below the critical value the vehicle is static. At the onset the net moment vanishes, which supplies *two* anchoring conditions,

$$\omega(t_{\text{crit}}) = 0, \qquad \dot{\omega}(t_{\text{crit}}) = 0. \tag{2}$$

Past the onset gravity feeds back positively: the linearized dynamics have *real* eigenvalues $\pm C_2$ and, with the measured moment ramp $M(t) = M_{\text{crit}} + \dot{M}\,\tau$ in onset-relative time $\tau = t - t_{\text{crit}}$, the exact solution of (1)–(2) collapses to the hyperbolic closed form

$$\boxed{\omega(\tau) = C_1\big(\cosh(C_2\tau) - 1\big), \qquad C_1 = K\dot{M}, \quad K = \frac{1}{W z_{\text{CoM}}}, \quad C_2 = \sqrt{\frac{W z_{\text{CoM}}}{J_P}}} \tag{3}$$

for $\tau \geq 0$ (and $\omega = 0$ for $\tau < 0$). Because $J_P C_2^2 = W z_{\text{CoM}}$, the amplitude C_1 is *fixed* by the measured ramp rate: with (C_2, K) pinned as per-configuration effective constants and the baseline fixed by continuity, **no shape parameter is free per run**. The time-quadratic model of the original submission is the $C_2 \to 0$ limit of (3) — it discards the gravity term that creates the instability, with 14–75% truncation error over the windows actually used — and is retained only as a baseline.

4 Signals and excitation window

Per run, the moment is reconstructed from the measured rotor speeds, $M(t) = C_T \sum_i b_{a,i}\,\Omega_i^2(t)$, and $\omega(t)$ is taken from the odometry rate. The window end is $i_{\text{end}} = \arg\max_t |M|$, *truncated at the first sample exceeding the allocator cap* M_{cap}: past it the thrust allocation saturates, the executed ramp is no longer linear, and the cosh family does not apply. (On the reference dataset the cap never binds; peak $|M| \leq 1.7$ N m.) The window start is the *last* sub-threshold sample before the peak, walking backwards — robust to spin-up blips that cross the threshold long before the actual ramp.

5 Onset-free run gates

Before any onset is estimated, each run must pass three gates evaluated on the measured moment trace alone (hence free of circularity):

1. **Ramp-slope error** $|\varepsilon| \leq 3\%$, with $\varepsilon = (|\dot{M}| - \dot{M}_{\text{cmd}})/\dot{M}_{\text{cmd}}$ and $\dot{M}$ the least-squares slope over the sub-segment $|M| \geq M_{\text{floor}}$ (0.7 N m roll, 0.4 N m pitch). The floors are the per-axis lower bounds of $|M_{\text{crit}}|$ over the ± 20 mm CoM-offset design box, so the gate scores the ramp exactly where an onset can occur: spin-up imperfections at near-zero moment cannot reject a run whose ramp is within spec where it matters. (On this dataset the floored gate passes all 140 runs; scored on the full window instead, 8 fast runs would be rejected by their early transients.) A 3% error in $\dot{M}$ perturbs $C_1 = K\dot{M}$ by 3% — about $7\times$ below the $\pm 20\%$ level the sensitivity analysis shows harmless.

2. **Window samples** $N_{\text{full}} \geq 38$: a necessary condition for any onset position. Because the window opens at $|M| = 10$ mN m while $M_{\text{crit}} \gtrsim 0.4$ N m, the onset cannot occur earlier than $\hat{N}_{\text{pre}} = M_{\text{crit}}^{\min}/(\dot{M}\,T_s) \geq 30$ samples into the window even at the fastest ramp; adding the $N_{\text{seg}} = 8$ post-onset fit minimum gives 38.

rate	$\lvert\varepsilon\rvert$ [%]		lin. RMSE [mN m]		N_{full}	N_{floor}
[N m/s]	med	max	med	max	min	min
0.10	0.14	1.16	2.3	3.8	484	102
0.20	0.19	1.55	1.9	3.3	257	71
0.30	0.17	0.83	1.8	2.5	191	56
0.45	0.15	2.21	1.7	2.8	130	52
0.65	0.23	0.90	1.8	2.9	95	47
0.90	0.39	2.40	1.9	2.9	77	33
1.20	0.86	1.80	2.3	4.2	53	31

Table 2: Per-rate summary of the gate quantities (20 runs per rate; slope and linearity on the floored segment). All values are inside the gates with wide margin; the mild growth of $\lvert\varepsilon\rvert$ toward 1.20 N m/s reflects the shorter averaging segment, not degraded execution (`analysis/gate_report.py`).

3. **Linearity RMSE** $\leq$ 30 mN m about the run's own best-fit line: a fault detector for aborted/stepped ramps, set at 10× the execution noise floor and above the healthy-run maximum.

The gates run *before* calibration so a poorly executed ramp cannot contaminate the dataset-coupled constraint $C_1 = K\dot{M}$. Figure 2 and Table 2 report the three gate quantities for all 140 runs, evaluated on the floored segment: the slope tracking error stays within ±2.4% (median 0.28%), the ramp-linearity RMSE within 4.2 mN m (median 2.0 — the execution noise floor itself, a factor 7 below the gate), and the shortest window carries 53 samples (1.4× the necessary minimum).

6 Onset detection

With (C_2, K) pinned, detection reduces to an exhaustive sweep of the single remaining unknown — the onset index. Each candidate is a plain arithmetic evaluation: no optimizer, no initial guess, no seed model.

Algorithm 1 COSH onset detection (single run)

Require: samples $\{(t_i, \omega_i)\}$ on the excitation window $[i_{\text{start}}, i_{\text{end}}]$; moment $M(t)$; calibrated (C_2, K); measured ramp slope $\dot{M}$; margin $N_{\text{seg}} = 8$
Ensure: onset index j^*; critical moment M_{crit}

1: $C_1 \leftarrow K\dot{M}$ ▷ onset conditions (2) fix the amplitude
2: **for** $j = i_{\text{start}} + N_{\text{seg}}, \ldots, i_{\text{end}} - N_{\text{seg}}$ **do** ▷ exhaustive sweep of the single unknown
3: $\quad C \leftarrow \text{median}\{\omega_i : i_{\text{start}} \leq i < j\}$ ▷ baseline by continuity
4: $\quad \text{cost}(j) \leftarrow \sum_{i=i_{\text{start}}}^{j-1} (\omega_i - C)^2 + \sum_{i=j}^{i_{\text{end}}} \big(\omega_i - [\,C_1(\cosh C_2(t_i - t_j) - 1) + C\,]\big)^2$
5: **end for**
6: $j^* \leftarrow \arg\min_j \text{cost}(j)$; $\quad M_{\text{crit}} \leftarrow M(t_{j^*})$

Two implementation details. *Baseline by median.* The pre-onset gyro trace is heavy-tailed (propeller vibration, spin-up spikes), and — more importantly — when a candidate j overshoots the true onset, the pre-segment tail contains the beginning of the hyperbolic rise. A mean baseline drifts with that leaked rise and partially cancels the residual it should expose, flattening the cost landscape; the median stays at the rest level until half the segment is contaminated, so misplaced candidates are penalized sharply and the minimum stays well defined. It is also a one-shot, seedless statistic, consistent with the no-optimizer design. An ablation rerunning the full pipeline — including the (C_2, K) calibration — with the mean instead (`analysis/baseline_stat_ablation.py`) leaves the run gating unchanged and 81/140 onsets identical, but biases the remainder toward *earlier* onsets, shrinking the directional means in magnitude by up to 30 mN m on the noisiest datasets. The route is indirect. At *fixed* (C_2, K) a baseline biased toward the motion direction *delays* the detected onset: the post-onset residual $C_1[\cosh(C_2(t - t_{\text{crit}}^{\text{real}})) - \cosh(C_2(t - t_{\text{crit}}^{\text{est}}))] - C$ can only vanish with a later split, and an oppositely biased baseline advances it. The dominant effect, however, is that the leaked-rise flattening of the cost landscape moves the CV-selected calibration to a different point on the (C_2, K) ridge — e.g. $C_2 : 8.00 \to 5.38$, $K : 0.060 \to 0.150$ on the most affected dataset — whose net effect is the earlier-onset magnitude shrink; the ramp-invariance CV itself is essentially unchanged. The pivot-free offset M_{ff} moves by at most 4.0 mN m (0.13 mm), well below the run-to-run scatter — the median removes a systematic bias and tunes nothing. A second ablation pins $C = 0$, the ideal-sensor value implied by the onset conditions: the measured pre-onset baseline is in fact nonzero (gyro bias plus residual spin-up motion; $|C|$ median 3.5, p90 17.6, max 47.8 mrad/s), yet the estimate barely moves (66/140 onsets identical, M_{ff} shift ≤ 4.9 mN m, i.e. 0.16 mm). The estimated baseline is therefore not load-bearing on this dataset; it is kept because it makes the sweep invariant to the gyro bias — a vehicle- and temperature-dependent quantity that need not stay this small — at zero parameter cost. Nor does C need the bias to be constant in time: it is re-estimated per run, so only *within-window* drift (windows last 1–10 s) matters. Measured over the guaranteed pre-onset segment ($|M| < 0.3$ N m, spans ≤ 3.0 s), the baseline wanders by 3.4 mrad/s (median; p90 24, max 68) — an upper bound that conflates true sensor drift with real elastic pre-onset motion (`analysis/bias_drift.py`). This wander is present in the very data on which the $C = 0$ bound and the rate-invariance certificates were earned, and its structure is covered by the perturbation absorption: the constant part goes to C, a τ-linear trend is absorbed by the span projection, and only the residual curvature reaches ρ. *Sweep margin.* The margin $N_{\text{seg}} = 8$, applied at both window ends, guarantees every candidate split leaves at least N_{seg} samples in each segment; it excises only degenerate candidates: near $j = i_{\text{start}}$ the baseline would be estimated from a handful of samples, and near $j = i_{\text{end}}$ the post-onset arc is too short for the cosh branch to be falsifiable (at $j = i_{\text{end}}$ its cost is nearly an empty sum). Since physically feasible onsets lie at least $\hat{N}_{\text{pre}} \geq 30$ samples inside the window (Sec. 5) and the window closes only after the response is well developed, the sweep remains exhaustive over every feasible onset.

Remark 1 *Because the model curve is fully determined before the post-onset data are seen, goodness of fit is a* prediction test*, not flexibility: across all gate-passing runs the normalized post-onset residual is $\approx 6\%$ (gyro-noise dominated) and invariant over the twelvefold ramp-rate range — the empirical certificate that the response stays in the cosh family.*

7 Calibration of the effective constants

Algorithm 1 presupposes (C_2, K). Although both have geometric expressions (3), they are *not* computed from geometry: z_{CoM} and especially J_P are not precisely known a priori, and — more importantly — the constants that make (3) reproduce the measured response are *effective* values that absorb the state-linear contributions of the unmodeled effects (the local slope of the gravity nonlinearity and the attitude gradient of the ground effect). Both are calibrated once per dataset.

Joint residual minimization is not a usable criterion in practice. On the fitted window the amplitude and the onset are nearly interchangeable under noise: a smaller amplitude with an earlier onset and a larger amplitude with a later onset produce post-onset curves that differ most near the onset itself, where the response is quadratic, $\frac{1}{2}C_1C_2^2\tau^2$, and of the order of the gyro noise. In the noiseless limit the post-onset shape determines all three parameters uniquely — the trade-off is an ill-conditioning statement, not a non-identifiability one — but under gyro noise the summed residual is nearly flat along this amplitude–onset direction: a *shallow ridge* in the (C_2, K) plane, observed directly on the data (the objective is piecewise-constant in the integer onset index, gradient-based and simplex optimizers stall on it, and a population-based global search costs about 4× more for no measurable gain). Per-run estimates — and with them the detected critical moments — scatter along the ridge. The calibration should therefore be read as *variance suppression*: choosing (C_2, K) once per dataset to minimize the dispersion of M_{crit} across ramp rates removes precisely that scatter, which is where the practical strength of the method lies.

The scatter direction is faint in the residual but loud in the deliverable: moving along the trade-off shifts each detected onset in *time*, and since $M_{\text{crit}} = M(t_{\text{on}})$ with M ramping at $\dot{M}$, an onset-time bias δt common across runs biases the critical moment by $\dot{M}\,\delta t$ — growing linearly with the ramp rate. A wrong pair thus *fans out* the identified M_{crit} across the twelvefold rate range, while the correct pair leaves it rate-invariant: exactly the direction a rate-consistency criterion sees best. We therefore calibrate by *physical self-consistency*: a static threshold must not depend on how fast it was approached. With runs at ramp rates $r \in \{0.10, \ldots, 1.20\}$ N m/s,

$$\text{score}(C_2, K) \;=\; \sum_{\text{dir}\in\{+,-\}} \text{CV}\Big(\big\{M_{\text{crit}}^{(r)}(C_2, K)\big\}_r\Big), \qquad \text{CV} = \text{std}/|\text{mean}|, \tag{4}$$

where $M_{\text{crit}}^{(r)}$ is the output of Algorithm 1 on the run with commanded rate r.

Search ranges. The stage-2 grid is centred on the stage-1 medians and spans $\pm 25\%$ in C_2 and $[-40\%, +40\%]$ in K, which is wide enough to absorb the scatter of the per-run fits (interquartile ranges of roughly $\pm 10\%$ in C_2) without reaching the unphysical extremes a fixed box admits. For reference, the earlier fixed box was $K \in [0.10, 0.50]$ $(\mathrm{N\,m})^{-1}$, i.e. $z_{\mathrm{CoM}} \in [0.06, 0.32]$ m at the vehicle weight, with the refinement allowed to move within the wider clip box $[0.05, 0.70]$ (z_{CoM} up to 0.63 m). For $C_2 \in [3, 8]$ rad/s, the implied pivot inertia $J_P = 1/(KC_2^2)$ spans roughly $[0.03, 1.1]$ $\mathrm{kg\,m^2}$, bracketing the parallel-axis estimate $J + mz_{\mathrm{CoM}}^2 \approx 0.2$–$0.35$ $\mathrm{kg\,m^2}$ by a factor of several on both sides.

Algorithm 2 Ramp-invariance consistency score

Require: set $\mathcal{G}$ of gated runs (both tip directions, spanning the rate set $\{0.10, \ldots, 1.20\}$ N m/s); candidate pair (C_2, K)

Ensure: consistency score s ▷ lower = M_{crit} more rate-invariant

```
for dir ∈ {+, −} do
    G_dir ← {ρ ∈ G : dir(ρ) = dir}
    for each run ρ ∈ G_dir do
        M_crit^(ρ) ← Algorithm 1 with (C2, K) pinned          ▷ stride-3 sweep
    end for
    CV_dir ← std({M_crit^(ρ)}_{ρ∈G_dir}) / |mean({M_crit^(ρ)}_{ρ∈G_dir})|
end for
return s ← CV+ + CV−   ▷ Eq. (4): a static threshold must not depend on the approach rate
```

Note that the ramp rate never appears explicitly in the score: each direction's gated runs span the full rate set, so the within-direction dispersion of the per-run critical moments *is* the rate dependence (plus repeatability scatter, which the score harmlessly co-penalizes). Should a rate be repeated, each run enters individually.

Algorithm 3 Effective-constant calibration (PNLS-located, then refined)

Require: gated runs of one dataset

Ensure: calibrated pair (C_2^*, K^*)

1. **locate (PNLS, runs first):** fit (C_1, C_2, C) per run by free nonlinear least squares; record C_2 and $K = C_1/\dot{M}$
2. $(C_2^0, K^0) \leftarrow$ per-dataset medians ▷ owes nothing to the ramp-invariance criterion
3. **refine:** evaluate Algorithm 2 on the 9×9 grid $C_2 \in C_2^0[0.75, 1.25]$, $K \in K^0[0.60, 1.40]$
4. $(C_2^*, K^*) \leftarrow \arg\min$ over that grid

Two-stage procedure. The two stages consume independent information. The PNLS pass reads the *shape* of each run's post-onset curve and is blind to how the threshold varies with ramp rate; the refinement reads exactly that variation and is nearly blind to the combination $KC_2^2 = 1/J_P$ that the shape fixes (Remark 2). Running PNLS first and refining about its medians keeps the search off the arbitrary edges of a fixed box: the refined constants land at $C_2 \in [3.52, 5.67]$ rad/s and $K \in [0.150, 0.338]$ $(\mathrm{N\,m})^{-1}$, whereas minimising the same score over the wide box $[3, 8] \times [0.05, 0.70]$ puts case $1/M_x$ on the upper bound at $C_2 = 8.000$ and spreads K over 0.060–0.520, a factor of 8.7. Accuracy and consistency are unaffected — RMS 1.65 mm against 1.64, median CV 2.5% against 2.5% (Tables 7 and 10) — so the gain is interpretability, not performance. Three of the twenty refined coordinates still sit on their local grid edge (case $3/M_x$ in C_2, cases 4 and $5/M_y$ in K), where the score genuinely prefers to move away from the shape-based centre.

Grid resolution. Algorithm 2 states the consistency criterion as a stand-alone routine — it wraps Algorithm 1 and returns the directional sum of coefficients of variation — and Algorithm 3 performs the search over it. Stage 2 evaluates (4) on a 9×9 uniform grid about the stage-1 centre, i.e. resolutions of about 0.13 rad/s in C_2 and $0.05\,K^0$ in K (81 evaluations per dataset). One pass suffices because the objective is flat along the (C_2, K) ridge; it is also *piecewise constant* (the onset is an integer index, so small parameter changes leave M_{crit} unchanged), which is why gradient-based or simplex optimizers stall on it, and a population-based global search (differential evolution) was found to cost about $4\times$ more for no measurable gain. During calibration the onset sweep of Algorithm 1 is subsampled by a stride of 3 samples for speed; the final identification runs at full resolution with the selected pair. Each evaluation is an arithmetic sweep, so no iterative optimizer is involved anywhere in the pipeline.

What the criterion buys. Table 3 reports the quantity (4) is built from: the coefficient of variation of $|M_{\text{crit}}|$ within a tip direction, over a rate sweep spanning a factor of 12.4 ($\dot{M} = 0.10$ to 1.24 N m/s, 14 runs per dataset, 140 runs in total). The calibrated constants hold the threshold to 2.4% in the mean and 4.2% at worst. The criterion is not flat in K: halving it about the CAD point nearly doubles the dispersion, to 7.9% in the mean and 14.6% at worst. Doubling it instead *improves* the CAD figures, to 3.1% and 4.8% — the calibration's own preference showing through, since every fitted K on the pitch axis exceeds the CAD value. That asymmetry is the disagreement of Table 4 seen from the objective's side.

Table 3: Coefficient of variation of $|M_{\text{crit}}|$ within a tip direction [%], averaged over the two directions. *Free*: (C_2, K) calibrated per dataset (twenty free numbers). *CAD*: constants derived from $z_{\text{CoM}} = 0.261$ m and the Table 5 inertias by the parallel-axis theorem, with nothing fitted. Last two columns: K mis-specified about the CAD point.

Case	Axis	Free	CAD	$K/2$	$2K$
1	M_x	4.2	5.2	8.7	4.7
1	M_y	2.5	6.9	14.6	2.9
2	M_x	0.9	2.7	5.8	2.0
2	M_y	1.4	5.6	10.3	2.2
3	M_x	1.8	1.9	4.6	3.0
3	M_y	1.8	3.2	7.5	2.2
4	M_x	2.4	2.6	4.7	3.2
4	M_y	2.9	4.6	8.1	3.2
5	M_x	2.6	3.5	6.4	3.0
5	M_y	3.8	5.6	9.7	3.9
mean		2.4	4.2	7.9	3.1
worst		4.2	6.9	14.6	4.8

Remark 2 (What the valley does and does not determine) *The objective* (4) *retains a shallow valley, and the two constants are* not *equally determined along it. Read literally,* $1/K$ *would be* $W z_{\text{CoM}}$*; across the ten datasets it implies* z_{CoM} *between* 0.061 *and* 0.554 *m, and the roll and pitch datasets of* the same airframe *disagree by up to a factor* 8 *(case 1:* 0.554 *versus* 0.068 *m). Since only horizontal masses are moved between configurations,* z_{CoM} *is common across them, so* $1/K$ *is better read as an effective gain than as a CoM height. The combination* $J_P = 1/(KC_2^2)$ *behaves quite differently:* 0.249 ± 0.016 *kg m*^{2} *over the five roll datasets* (CV = 6.5%) *and* 0.140 ± 0.034 *over the five pitch datasets, against* CV = 65% *and* 49% *for* $1/K$.

Table 4 makes the comparison explicit against a fully independent reference. The CAD model gives $z_{\text{CoM}} = 0.261$ *m — measured from the landing-gear contact plane, so it is already referred to the pivot and needs no datum correction — together with the CoM inertias of Table 5, so the parallel-axis theorem fixes the pivot inertia without any fit,* $J_P = J_{\text{CAD}} + m(z_{\text{CoM}}^2 + l_p^2) = 0.334$ *and* 0.309 *kg m*^{2}*, and with it both constants:* $C_2 = 4.97$ *and* 5.17 *rad/s,* $K = 0.121$ *(N m)*$^{-1}$*. The products of inertia are four orders of magnitude below the diagonal, so treating the two axes independently is justified.*

The pivot arm entering that expression is measured, not assumed. The mocap marker traces an arc about the ground contact during the tip-over, and `estimate_pivot_from_mocap` *fits it with the circle centre held at ground level, leaving the horizontal offset and the radius free. Over the* 138 *runs with a valid arc the fit returns* $l_p = 0.1404 \pm 0.0036$ *m on roll and* 0.1126 ± 0.0066 *m on pitch, against the* 0.140 *and* 0.110 *m used throughout* (-0.3% *and* -2.3%)*, with a circle residual of* 0.1–0.2 *mm. The same fit puts the marker at* 0.3172 ± 0.0012 *m above the pivot, matching the* $\sqrt{R^2 - l_p^2} = 0.3167$ *m implied by the radius and the rotor-hub height* $h = 0.315$ *m used in the ground-effect model. The geometry is therefore consistent to about a millimetre: the mocap heights are referred to the ground contact, the CAD height is referred to the landing gear, and the CAD CoM lands* 56 *mm below the marker plane, as it should for an airframe carrying its battery low. Both geometric inputs to the parallel-axis floor are thus independently confirmed, and the identified* J_P *is the only quantity left that can account for the gap.*

The disagreement is not spread evenly over the (C_2, K) *plane, and that is the substance of the remark. The constant the deliverable is sensitive to,* C_2*, agrees: the per-run nonlinear least squares gives a median of* 4.765 *rad/s,* 4.2% *and* 7.8% *below the CAD values, and the calibrated means are within* 14%*. The scale of the pair does not: read literally the calibration puts* J_P *at* 0.249 *and* 0.140 *kg m*^{2}*, which are* 25% *and* 55% *below CAD and — more tellingly —* 12% *and* 46% *below the parallel-axis floor* $m(z_{\text{CoM}}^2 + l_p^2)$ *obtained by setting* J_{CAD} *to zero.* The identified inertia is not physically attainable. *Since* $C_2^2 = W z_{\text{CoM}}/J_P$ *is* *confirmed while J_P is not, what the calibration fails to resolve is the scale of the pair, not their ratio — exactly the K direction that the log sensitivities below identify as the weak one.*

Table 4: Calibrated constants against the CAD reference ($z_{\mathrm{CoM}} = 0.261$ m, Table 5 inertias, parallel-axis theorem; nothing fitted). CV and spread are over the five datasets of the axis. The last row of each block also gives the parallel-axis floor, the value J_P would take if the CoM inertia were zero.

Axis	Quantity	mean	CV	spread	CAD	difference
Roll	C_2 [rad/s]	5.65	34.8%	2.29×	4.97	+13.6%
	$1/K = Wz_{\mathrm{CoM}}$ [N m]	8.71	64.7%	5.00×	8.25	+5.6%
	$J_P = 1/(KC_2^2)$ [kg m^2]	0.249	6.5%	1.16×	0.334	−25.3%
	(floor 0.283)					−11.8%
Pitch	C_2 [rad/s]	4.90	24.5%	1.71×	5.17	−5.2%
	$1/K = Wz_{\mathrm{CoM}}$ [N m]	3.46	49.2%	2.89×	8.25	−58.0%
	$J_P = 1/(KC_2^2)$ [kg m^2]	0.140	24.1%	1.95×	0.309	−54.7%
	(floor 0.258)					−45.8%

Pitch is the weaker axis throughout, and its error is systematic rather than scattered: all five pitch datasets put $1/K$ below the CAD value (1.92 to 5.56 against 8.25 N m), whereas the roll datasets bracket it. The excitation window is shortest on that axis — τ_{end} has a median of 0.450 s against 0.555 s for roll, and falls to 0.23 s at the highest rates — so $x = C_2\tau_{\mathrm{end}}$ is smallest there and the curvature information that separates J_P from Wz_{CoM} is correspondingly weakest.

This is structural, not accidental. At the window edge, with $x = C_2\tau_{\mathrm{end}}$, the log sensitivities are

$$\frac{\partial \ln \omega}{\partial \ln C_2} = x \coth\frac{x}{2} \;\geq\; 2, \qquad \frac{\partial \ln \omega}{\partial \ln K} = 1,$$

so the two cancel along $\delta \ln K = -x\coth(x/2)\,\delta \ln C_2$, which for small x is exactly $\delta \ln K = -2\,\delta \ln C_2$, i.e. $KC_2^2 = 1/J_P$ constant. The valley therefore runs along the direction in which C_2 carries twice the leverage of K; equivalently, near the onset $\omega \simeq (\dot{M}/J_P)\tau^2/2$ depends on the pair only through J_P. The measured spread confirms the factor: over the search box $1/K$ moves by $2.36\times$ while C_2 moves by $1.54\times$, and $\ln 2.36/\ln 1.54 = 1.99$.

Robustness certificate: three constants for ten datasets. Because the individual constants are weakly determined, it is reasonable to ask whether (C_2, K) are doing the work of a two-parameter curve family rather than of the geometry they name. A direct test is to remove the freedom. Physics requires z_{CoM} to be shared by the whole vehicle and J_P to be shared by all datasets on a given axis, so we reparametrize by

$$\big(Wz_{\mathrm{CoM}},\ J_{P,\mathrm{roll}},\ J_{P,\mathrm{pitch}}\big), \qquad C_2 = \sqrt{\frac{Wz_{\mathrm{CoM}}}{J_{P,\mathrm{axis}}}}, \qquad K = \frac{1}{Wz_{\mathrm{CoM}}}, \tag{5}$$

minimize the *same* score (4) summed over all ten datasets, and rerun the identification. Twenty free numbers become three; the roll and pitch inertias decouple, so the inner search is two independent scalar minimizations per candidate Wz_{CoM}. The optimum is $Wz_{\mathrm{CoM}} = 5.50$ N m ($z_{\mathrm{CoM}} = 0.174$ m), $J_{P,\mathrm{roll}} = 0.240$ and $J_{P,\mathrm{pitch}} = 0.153$ kg m^2, giving $C_2 = 4.79$ and 6.00 rad/s and $K = 0.182$ (N m)$^{-1}$ — close to the representative values quoted throughout. Table 5 compares the delivered CoM offsets.

Collapsing the twenty free numbers to three changes the RMS error by 0.03 mm and no individual case by more than 0.11 mm. The identical mean error, −0.95 mm in both columns, further shows that the residual bias is a property of the vehicle–contact geometry, common to every estimator, and not of the calibration. In the constrained form C_2 is also derived rather than searched, so it cannot settle on a search bound.

One caution completes the picture: this does *not* measure z_{CoM}. The score varies by only 3.6% across the whole box $z_{\mathrm{CoM}} \in [0.15, 0.35]$ m, so every physically plausible height performs comparably. The identification therefore does not depend on knowing z_{CoM}, which is also why the individual constants are best reported as effective values rather than as geometry.

The same test with no fitted number at all. The three-constant form above still minimizes (4). The strictest variant fits nothing: CAD supplies z_{CoM} and the CoM inertias, the parallel-axis theorem supplies J_P, and both constants follow. Table 6 reruns the identification on that point.

Table 5: Identified CoM offset [mm] against the load-cell truth, with the constants calibrated per dataset (twenty free numbers) and with the physically shared parametrization (5) (three).

Case	Axis	Truth	Free	error	Constrained	error
1	M_x	−2.90	−1.73	+1.17	−1.64	+1.26
1	M_y	−11.45	−11.13	+0.32	−11.25	+0.20
2	M_x	−14.29	−15.17	−0.88	−15.16	−0.87
2	M_y	−9.90	−10.49	−0.59	−10.53	−0.63
3	M_x	−5.26	−8.62	−3.36	−8.66	−3.40
3	M_y	3.14	0.73	−2.41	0.63	−2.51
4	M_x	6.67	5.26	−1.41	5.23	−1.44
4	M_y	2.40	0.16	−2.24	0.11	−2.29
5	M_x	10.91	10.70	−0.21	10.79	−0.12
5	M_y	−10.89	−10.73	+0.16	−10.62	+0.27
RMS error				1.64		1.67
max |error|				3.36		3.40
mean error				−0.95		−0.95

Table 6: Identified CoM offset [mm] with the constants taken entirely from CAD — $z_{\mathrm{CoM}} = 0.261$ m, Table 5 inertias, no quantity fitted to anything. *CV*: rate consistency of $|M_{\mathrm{crit}}|$ as in Table 3.

Case	Axis	J_P	C_2	K	CV [%]	Truth	Identified (error)
1	M_x	0.320	4.95	0.127	5.2	−2.90	−1.61 (+1.29)
1	M_y	0.297	5.15	0.127	6.9	−11.45	−11.16 (+0.29)
2	M_x	0.334	4.97	0.121	2.7	−14.29	−15.16 (−0.87)
2	M_y	0.309	5.17	0.121	5.6	−9.90	−10.37 (−0.47)
3	M_x	0.334	4.97	0.121	1.9	−5.26	−8.68 (−3.42)
3	M_y	0.309	5.17	0.121	3.2	3.14	0.74 (−2.40)
4	M_x	0.334	4.97	0.121	2.6	6.67	5.24 (−1.43)
4	M_y	0.309	5.17	0.121	4.6	2.40	0.34 (−2.06)
5	M_x	0.334	4.97	0.121	3.5	10.91	10.78 (−0.13)
5	M_y	0.309	5.17	0.121	5.6	−10.89	−10.34 (+0.55)
CAD only (nothing fitted)						RMS 1.64, max 3.42, mean −0.87	
Free calibration (twenty fitted numbers)						RMS 1.64, max 3.36, mean −0.95	

The RMS error is 1.64 mm either way. Twenty fitted numbers and none deliver the same offset, and the mean error stays near −0.9 mm in both — a residual belonging to the vehicle–contact geometry, not to the calibration. What the fitting does buy is rate consistency: the CV rises from 2.4% to 4.2% in the mean and from 4.2% to 6.9% at worst when the constants are taken from CAD.

This is the sharpest available statement of what the constants are for. The calibration cannot be measuring the rig geometry — its J_P is below the parallel-axis floor — yet the geometry it fails to recover can be substituted wholesale without moving the deliverable. Both facts follow from the same structure: the identified threshold responds to C_2, which the two routes agree on, and only weakly to K, which is where they part.

8 Aggregation and deliverables

Directional means over the gated runs give $\bar{M}_+$ and $\bar{M}_-$. The take-off feed-forward moment is the *pivot-free average*

$$M_{\mathrm{ff}} = \tfrac{1}{2}\,(\bar{M}_+ + \bar{M}_-), \tag{6}$$

in which the pivot-arm terms $\pm(W-f)\,l_p$ — and any thrust-proportional contamination acting through the arm l_p, ground effect included — cancel by antisymmetry, while landing-gear asymmetry is deliberately *retained*: the same contact-phase moment acts at lift-off, where M_{ff} compensates it. The CoM offset follows from the moment-to-offset conversion of M_{ff} and is used for validation against load-cell ground truth only; M_{ff} itself is the quantity applied in flight. A roll excitation (M_x) senses the y-offset via $M_{\mathrm{ff},x} = +W y_{\mathrm{off}}$; a pitch excitation ($M_y$) senses the x-offset via $M_{\mathrm{ff},y} = -W x_{\mathrm{off}}$.

case	axis	λ_{truth} [mm]	estimated CoM offset $\hat{\lambda} \pm$ CI$_{95}$ [mm] COSH	COSH$_{\text{wide}}$	COSH$_{\text{CAD}}$	NLS	CPD$_{\text{N}}$	CPD$_{\text{R}}$	CUSUM
1	x	−2.90	−1.65 ±0.94	−1.73 ±0.93	−1.61 ±1.02	−2.05 ±1.30	−1.46 ±1.03	−1.20 ±1.51	−1.15 ±1.24
1	y	−11.45	−11.18 ±0.42	−11.13 ±0.43	−11.16 ±0.78	−11.35 ±1.37	−11.21 ±0.61	−10.79 ±0.80	−9.59 ±3.88
2	x	−14.29	−15.15 ±0.36	−15.17 ±0.36	−15.16 ±0.64	−15.03 ±0.76	−15.09 ±0.70	−14.10 ±4.15	−15.00 ±0.52
2	y	−9.90	−10.57 ±0.41	−10.49 ±0.42	−10.37 ±0.80	−9.60 ±1.13	−9.91 ±0.52	−9.61 ±0.46	−8.54 ±4.02
3	x	−5.26	−8.60 ±0.45	−8.62 ±0.46	−8.68 ±0.48	−8.54 ±0.74	−8.11 ±1.20	−4.63 ±4.72	−8.45 ±0.92
3	y	+3.14	+0.72 ±0.40	+0.73 ±0.39	+0.74 ±0.60	+1.06 ±0.97	+0.71 ±0.82	+0.94 ±1.00	+1.12 ±3.39
4	x	+6.67	+5.24 ±0.65	+5.26 ±0.65	+5.24 ±0.66	+4.94 ±0.88	+5.57 ±0.72	+3.58 ±2.19	+5.01 ±0.91
4	y	+2.40	+0.14 ±0.61	+0.16 ±0.61	+0.34 ±0.79	−0.64 ±1.49	−0.66 ±1.39	−0.60 ±1.36	−0.22 ±1.36
5	x	+10.91	+10.78 ±0.73	+10.70 ±0.69	+10.78 ±0.80	+10.05 ±1.10	+10.29 ±0.65	+11.16 ±3.75	+10.25 ±0.74
5	y	−10.89	−10.57 ±0.68	−10.73 ±0.70	−10.34 ±0.93	−11.11 ±1.68	−10.84 ±0.98	−10.95 ±0.95	−10.84 ±1.00
component RMS [mm]			1.65	1.64	1.64	1.72	1.67	1.65	1.82
max |component| [mm]			3.34	3.36	3.42	3.28	3.06	3.09	3.19
mean component [mm]			−0.93	−0.95	−0.87	−1.07	−0.91	−0.46	−0.58
radial RMS [mm]			2.33	2.31	2.32	2.43	2.36	2.34	2.57
radial max [mm]			4.12	4.14	4.18	3.89	3.74	4.31	3.77

Table 7: Identified CoM offset against the load-cell truth, on identical inputs (140 gated runs). **COSH** is the reported estimator, its constants calibrated in a neighbourhood of the per-run PNLS medians; COSH$_{\text{wide}}$ minimises the same score over the box $[3,8] \times [0.05, 0.70]$; COSH$_{\text{CAD}}$ derives the constants from CAD geometry with nothing fitted. A roll excitation (M_x) senses y_{off} and a pitch excitation (M_y) senses x_{off}, so each case contributes both components; the *radial* rows combine them per case as $\sqrt{e_x^2 + e_y^2}$.

9 Comparison with per-run estimators: NLS, CPD, CUSUM

Six alternatives were run on *identical* inputs — same signals, excitation windows, allocator caps and onset-free gates (the gated run set is method-independent because the gates use the moment trace only), 140 runs in total (`analysis/nls_comparison.py`):

- **COSH$_{\text{wide}}$**, **COSH$_{\text{CAD}}$** — the same detector with the constants chosen differently: the score minimised over the wide box $[3,8]\times[0.05, 0.70]$ instead of around the PNLS centre, and the constants derived from CAD geometry with nothing fitted (Sec. 7);
- **NLS** — the per-run stage-1 fit used on its own: nonlinear least squares (SCIPY `least_squares`, trust-region reflective) for free (C_1, C_2, C), with C_2 bounded to $[3, 8]$ rad/s and the onset swept locally around a quadratic-model seed;
- **CPD$_{\text{N}}$ / CPD$_{\text{R}}$** — optimal single change-point detection (`ruptures`; with one break the search over all admissible splits is exact) under the Gaussian mean+variance cost and the nonparametric RBF-kernel cost, respectively;
- **CUSUM** — the tabular sequential detector with allowance $k = 2\sigma$ against the pre-onset vibration.

Figure 3 is the readable summary: per case–axis pair it shows six estimators' CoM offset with its 95% confidence interval (Welch t on the pivot-free average, run-to-run scatter) against the load-cell truth. The full data sit in the tables: Table 7 compares the deliverable — the CoM offset from the pivot-free average (6) — against the load-cell ground truth; Table 8 gives the feed-forward compensation moment itself; Table 9 lists the underlying directional critical-moment means, pivot-free offsets and their confidence intervals; and Table 10 compares the ramp-rate dispersion of the directional critical moments.

Three observations. *(i) The deliverable is detector-agnostic:* against the load-cell truth all five estimators land at RMS 1.6–1.8 mm per component, equivalently 2.3–2.6 mm radial (Table 7, bottom rows) — the pivot-free average (6) cancels each method's largely common-mode onset bias between the + and − directions, and 5–7 runs per direction average the rest. That the final CoM offset does not depend on the detection methodology is itself a validation certificate — and Fig. 4 sharpens it: the residual errors that do remain (cases 3–4) are shared by all five estimators, placing them in the vehicle–contact geometry rather than in any onset detector. *(ii) Run-level robustness is where the methods separate:* COSH has the lowest CV in 16/20 direction groups (CPD$_{\text{N}}$ takes 3, CPD$_{\text{R}}$ 1), median 2.5% against 4.1–7.1%, and its worst group is 5.7% where the alternatives reach 8.6–27.4% — the nonparametric detectors (CPD$_{\text{R}}$, CUSUM) occasionally misplace the onset badly on short fast-ramp windows. Equivalently, the 95% confidence interval on the CoM offset is ±0.3–0.9 mm under COSH — the narrowest in 9 of 10 case–axis pairs

		truth	feed-forward compensation moment $M_{\text{ff}} \pm$ CI$_{95}$ [mN m]						
case	axis	[mN m]	COSH	COSH$_{\text{wide}}$	COSH$_{\text{CAD}}$	NLS	CPD$_{\text{N}}$	CPD$_{\text{R}}$	CUSUM
1	x	−87.2	-49.6 ± 28.3	-52.2 ± 28.1	-49.1 ± 30.8	-61.7 ± 39.1	-43.9 ± 30.9	-36.1 ± 45.4	-34.5 ± 37.2
1	y	+344.4	$+336.3 \pm 12.6$	$+334.8 \pm 12.8$	$+334.5 \pm 23.5$	$+341.4 \pm 41.1$	$+337.0 \pm 18.2$	$+324.6 \pm 24.2$	$+288.6 \pm 116.8$
2	x	−451.4	-478.6 ± 11.5	-479.3 ± 11.3	-479.4 ± 20.3	-474.8 ± 23.9	-476.8 ± 22.0	-445.4 ± 131.1	-473.7 ± 16.6
2	y	+312.7	$+334.0 \pm 13.0$	$+331.4 \pm 13.4$	$+327.4 \pm 25.4$	$+303.3 \pm 35.8$	$+313.0 \pm 16.6$	$+303.7 \pm 14.6$	$+269.8 \pm 126.9$
3	x	−166.2	-271.7 ± 14.3	-272.4 ± 14.5	-274.0 ± 15.2	-269.9 ± 23.5	-256.0 ± 37.8	-146.3 ± 149.0	-266.8 ± 29.2
3	y	−99.2	-22.8 ± 12.5	-23.1 ± 12.3	-23.0 ± 18.8	-33.5 ± 30.7	-22.4 ± 25.8	-29.6 ± 31.5	-35.3 ± 107.2
4	x	+210.7	$+165.5 \pm 20.6$	$+166.1 \pm 20.6$	$+165.6 \pm 20.9$	$+155.9 \pm 27.9$	$+176.0 \pm 22.6$	$+113.2 \pm 69.0$	$+158.3 \pm 28.7$
4	y	−75.8	-4.3 ± 19.3	-5.1 ± 19.4	-10.3 ± 25.0	$+20.1 \pm 47.2$	$+21.0 \pm 43.9$	$+19.0 \pm 43.1$	$+6.9 \pm 43.1$
5	x	+344.6	$+340.7 \pm 22.9$	$+338.1 \pm 21.9$	$+340.7 \pm 25.4$	$+317.4 \pm 34.7$	$+325.1 \pm 20.7$	$+352.5 \pm 118.3$	$+323.8 \pm 23.5$
5	y	+344.0	$+333.8 \pm 21.5$	$+338.9 \pm 22.0$	$+325.5 \pm 29.4$	$+351.0 \pm 53.0$	$+342.3 \pm 30.8$	$+346.0 \pm 30.1$	$+342.3 \pm 31.6$

Table 8: The quantity actually compensated, $M_{\text{ff}} = \pm\frac{1}{2}(M_+ + M_-)$, for the same runs. Changing the COSH constants moves it by at most 5.1 mN m (PNLS-centred versus wide box) and 13.4 mN m (CAD versus wide box), against magnitudes up to 479 mN m — the compensation is insensitive to which of the three constant choices is adopted.

— against up to ±4–5 mm for the nonparametric detectors (Fig. 3). The paired per-run critical moments (NLS vs. COSH) differ by 33 mN m at the median but up to 269 mN m — the ridge of Sec. 7 realized run by run. *(iii) The biases carry signatures:* NLS sits 3.7% *low* in mean $|M|$ (the early-onset drift of the free amplitude–onset fit), while CPD$_{\text{N}}$ and CUSUM sit 3.0% and 3.1% *high* — detection lag, since a distribution change must accumulate post-onset evidence before it is declared; both signatures largely cancel in (6). COSH needs no per-run tuning (CUSUM's allowance k, CPD's minimum segment) and a *single* run already sits within a few percent of its directional mean — which is what enables run-count reduction and per-run gating diagnostics.

10 A bound on the ground-contact budget

This section bounds the near-ground contribution to the directional thresholds; it does not validate a particular aerodynamic model, and the closing paragraphs say why it cannot. The static tip-over thresholds tie each directional critical moment to independently measured quantities: $M_\pm = \pm(W - f)\, l_{\text{arm}} + W y_{\text{off}}$ (roll) and $M_\pm = \pm(W - f)\, l_{\text{arm}} - W x_{\text{off}}$ (pitch), with W and the offsets pinned by the load-cell truth, f the measured collective thrust at the onset, and the pivot arms fitted from the *position* trace: a circle fit of the odometry motion about the contact line per run (`estimate_pivot_from_mocap`); each run is predicted with its own arm and onset thrust, and the per-run predictions and residuals are then averaged per case/axis/direction. The kinematic arms come out remarkably consistent — 140.4 ± 2.3 mm (roll) and 112.7 ± 6.6 mm (pitch) across the five configurations, matching the geometric values (140/110 mm). Every ingredient is independent of the identified moments, so this is a genuine forward test — and no simulation enters anywhere: measured trajectories are combined with closed-form coefficients evaluated at the measured geometry.

Ground effect enters through *both* channels of the pivot-moment decomposition $\Delta M_{\text{GE}} = \text{sgn} \cdot a + b\, M$ (`analysis/ge_linearity.py`): the thrust channel $a = c_a f l$ (rotor-common surplus plus the rotor-differential moment, arm exactly l) and the moment-proportional channel b. The onset balance $M(1 + b) + \text{sgn}\, a = T_{\text{rigid}}$ gives

$$M_{\text{pred}} = \frac{T_{\text{rigid}} - \text{sgn} \cdot c_a f l}{1 + b}, \qquad T_{\text{rigid}} = \text{sgn}\,(W - f)\, l + S_{\text{off}}, \tag{7}$$

evaluated at three levels of physics: *(a)* no ground effect; *(b)* ground effect *without* rotor–rotor interference — the per-rotor Cheeseman–Bennett superposition at the tilted heights ($c_a = 1.03\%$, $b = 1.03\%$); and *(c)* ground effect *with* rotor interference — the image superposition over the rotor-centre distances, $s_{\text{rot},j} = (R^2/4) \sum_n (z_j{+}z_n)/[d_{jn}^2 + (z_j{+}z_n)^2]^{3/2}$, $\gamma_j = (1 - s_{\text{rot},j})^{-1}$, whose self term reduces exactly to Cheeseman–Bennett and which contains *no empirical constant of any kind* ($c_a = b = 4.31\%$).

Three observations. *(i) A systematic deficit exists, and it is physical.* The rigid no-GE prediction over-predicts the threshold magnitude in all 20 direction groups, by 199 mN m on average — a strictly one-sided gap, far above the identification scatter and shared by all five estimators (last column of Table 12), hence a property of the contact physics, not of the onset detector. *(ii) Its size is what near-ground aerodynamics predicts.* The parameter-free interference model — evaluated at the measured geometry,

method		1x	1y	2x	2y	3x	3y	4x	4y	5x	5y
COSH	$\bar{M}_-$	−1.043	−0.423	−1.824	−0.675	−1.538	−0.983	−1.138	−0.999	−1.016	−0.633
	$\bar{M}_+$	+0.944	+1.096	+0.867	+1.343	+0.995	+0.938	+1.469	+0.991	+1.698	+1.301
	M_{ff}	−0.050	+0.336	−0.479	+0.334	−0.272	−0.023	+0.165	−0.004	+0.341	+0.334
	CI_{95}	±0.028	±0.013	±0.011	±0.013	±0.014	±0.013	±0.021	±0.019	±0.023	±0.022
$\mathrm{COSH}_{\mathrm{wide}}$	$\bar{M}_-$	−1.054	−0.421	−1.829	−0.675	−1.542	−0.983	−1.136	−0.998	−1.019	−0.645
	$\bar{M}_+$	+0.950	+1.090	+0.870	+1.338	+0.997	+0.937	+1.468	+0.988	+1.695	+1.323
	M_{ff}	−0.052	+0.335	−0.479	+0.331	−0.272	−0.023	+0.166	−0.005	+0.338	+0.339
	CI_{95}	±0.028	±0.013	±0.011	±0.013	±0.014	±0.012	±0.021	±0.019	±0.022	±0.022
$\mathrm{COSH}_{\mathrm{CAD}}$	$\bar{M}_-$	−0.992	−0.372	−1.787	−0.597	−1.552	−0.959	−1.128	−0.949	−0.963	−0.584
	$\bar{M}_+$	+0.894	+1.041	+0.828	+1.252	+1.004	+0.913	+1.459	+0.928	+1.644	+1.235
	M_{ff}	−0.049	+0.334	−0.479	+0.327	−0.274	−0.023	+0.166	−0.010	+0.341	+0.326
	CI_{95}	±0.031	±0.024	±0.020	±0.025	±0.015	±0.019	±0.021	±0.025	±0.025	±0.029
NLS	$\bar{M}_-$	−1.018	−0.373	−1.792	−0.641	−1.573	−0.990	−1.151	−0.928	−0.989	−0.539
	$\bar{M}_+$	+0.894	+1.056	+0.843	+1.248	+1.033	+0.923	+1.463	+0.969	+1.624	+1.241
	M_{ff}	−0.062	+0.341	−0.475	+0.303	−0.270	−0.033	+0.156	+0.020	+0.317	+0.351
	CI_{95}	±0.039	±0.041	±0.024	±0.036	±0.024	±0.031	±0.028	±0.047	±0.035	±0.053
$\mathrm{CPD}_{\mathrm{N}}$	$\bar{M}_-$	−1.048	−0.456	−1.852	−0.707	−1.573	−1.059	−1.164	−1.005	−1.033	−0.647
	$\bar{M}_+$	+0.960	+1.130	+0.898	+1.333	+1.061	+1.014	+1.516	+1.047	+1.683	+1.332
	M_{ff}	−0.044	+0.337	−0.477	+0.313	−0.256	−0.022	+0.176	+0.021	+0.325	+0.342
	CI_{95}	±0.031	±0.018	±0.022	±0.017	±0.038	±0.026	±0.023	±0.044	±0.021	±0.031
$\mathrm{CPD}_{\mathrm{R}}$	$\bar{M}_-$	−1.023	−0.455	−1.680	−0.706	−1.339	−1.055	−1.179	−0.986	−0.932	−0.637
	$\bar{M}_+$	+0.950	+1.105	+0.789	+1.314	+1.046	+0.995	+1.406	+1.024	+1.637	+1.328
	M_{ff}	−0.036	+0.325	−0.445	+0.304	−0.146	−0.030	+0.113	+0.019	+0.353	+0.346
	CI_{95}	±0.045	±0.024	±0.131	±0.015	±0.149	±0.032	±0.069	±0.043	±0.118	±0.030
CUSUM	$\bar{M}_-$	−1.046	−0.476	−1.861	−0.714	−1.608	−1.029	−1.181	−1.044	−1.048	−0.667
	$\bar{M}_+$	+0.977	+1.054	+0.913	+1.254	+1.075	+0.958	+1.498	+1.058	+1.695	+1.351
	M_{ff}	−0.035	+0.289	−0.474	+0.270	−0.267	−0.035	+0.158	+0.007	+0.324	+0.342
	CI_{95}	±0.037	±0.117	±0.017	±0.127	±0.029	±0.107	±0.029	±0.043	±0.024	±0.032

Table 9: Directional critical-moment means $\bar{M}_\pm$, pivot-free offset M_{ff} and its 95% confidence interval (Welch t) [N m] per estimator. Columns are case–axis pairs (e.g. 1x = case 1, roll). The three COSH variants track each other closely, while NLS magnitudes sit low (early onsets) and $\mathrm{CPD}_{\mathrm{N}}$/CUSUM high (detection lag); M_{ff} stays nearly method-invariant throughout.

driven by the measured thrust, with no tuned constant — accounts for 82% of it, leaving a mean deficit of −36 mN m and breaking the one-sidedness (12/20); neglecting rotor–rotor interference accounts for only a fifth (−159 mN m, still 20/20 one-sided), so interference is not negligible at this clearance.

What it does not do is close the gap, and the relative figures are the honest way to say so. Against a median threshold of 1008 mN m the three levels leave 18.4%, 14.5% and 5.2% in the median, and 49.0%, 41.9% and 20.2% at their worst group. So the best available model, evaluated with no tuned constant, still misses each directional threshold by about a twentieth of its value, and by a fifth in the worst case. That disagreement is not the onset detector: the four alternative estimators bracket the identified moment within a median 108 mN m (last column of Table 12), and the interference prediction lands inside that bracket in 9 of the 20 groups where the other two levels never do. It is a property of the contact physics, and the rest of this section is about how far it can be pinned down — which is not far.

(iii) The check cannot attribute the deficit uniquely, and the argument does not need it to. A contact-geometry explanation fits equally well: if the normal-force resultant acts a distance δl *inboard* of the kinematic pivot circle — as a finite, compliant contact patch implies — the least-squares fit gives $\delta l = 19.2 \pm 1.8$ mm with a residual RMS of 82.3 mN m, statistically indistinguishable from the 86.0 mN m of the ground-effect hypothesis (`analysis/static_attribution.py`). The two are degenerate because both act as force × length in the constant part; the only discriminating structure is the moment-proportional channel $b\,M$, whose signal across the groups (18–79 mN m over a 4.3× range of M_{crit}) sits below the group-to-group scatter of the deficit (77 mN m), so a free fit returns $b = -2.9 \pm 5.2\%$ — consistent with the model value +4.31% and with zero alike. This check therefore *bounds the combined ground-contact budget* at ∼0.2 N m per direction and confirms it has the magnitude and sign that ground effect predicts; it is not an identification of ground effect against contact compliance.

The floor is set by the contact arm itself. The threshold's sensitivity to it is $\mathrm{d}M/\mathrm{d}l = W - f = 10.6$ N, so the 61 mN m median residual that survives the interference model is 5.8 mm of arm — the same order as the run-to-run spread of the circle fit inside a single group (0.7–8.4 mm, median 1.5), and smaller

case	axis	COSH		$\mathrm{COSH}_{\mathrm{wide}}$		$\mathrm{COSH}_{\mathrm{CAD}}$		NLS		$\mathrm{CPD}_{\mathrm{N}}$		$\mathrm{CPD}_{\mathrm{R}}$		CUSUM	
		CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+
1	x	3.9	5.8	**3.8**	5.7	5.6	**5.5**	6.4	7.7	4.4	6.1	7.3	8.6	5.9	6.7
1	y	4.4	**2.2**	**4.2**	2.3	12.3	3.2	17.1	7.2	5.0	3.3	7.0	4.4	4.8	24.0
2	x	1.3	**1.6**	**1.2**	1.6	2.3	2.9	2.5	4.2	2.5	2.2	15.5	22.1	1.8	2.4
2	y	1.6	2.1	**1.5**	2.1	8.1	3.0	7.1	5.8	3.1	2.5	3.3	**2.0**	3.0	21.9
3	x	**1.7**	**2.3**	1.7	2.4	1.7	2.5	2.4	4.1	4.9	4.2	23.9	8.5	3.1	4.7
3	y	**1.7**	2.6	1.7	**2.6**	2.9	3.9	3.2	7.0	4.4	4.1	4.4	6.0	12.4	22.8
4	x	3.3	2.2	3.4	2.2	**3.1**	2.5	3.8	3.5	4.0	**1.6**	5.6	10.3	3.9	3.5
4	y	**2.8**	**3.7**	2.9	3.7	4.3	4.8	10.3	6.1	7.5	7.2	7.8	6.9	6.7	7.3
5	x	3.8	2.4	3.9	2.1	5.6	**1.5**	7.2	2.6	**3.6**	2.0	27.4	3.0	4.0	2.3
5	y	5.3	**3.1**	**5.3**	3.1	7.1	4.6	19.8	5.4	8.6	3.8	8.4	3.8	8.3	4.0
median / worst [%]		2.5 / 5.8		2.5 / 5.7		3.5 / 12.3		5.9 / 19.8		4.1 / 8.6		7.1 / 27.4		4.7 / 24.0	

Table 10: Ramp-rate dispersion (CV [%]) of the directional critical moments. Bold marks the lowest CV in each of the 20 direction groups: COSH in 8, $\mathrm{COSH}_{\mathrm{wide}}$ in 6, $\mathrm{COSH}_{\mathrm{CAD}}$ in 3, $\mathrm{CPD}_{\mathrm{N}}$ in 2, $\mathrm{CPD}_{\mathrm{R}}$ in 1. The underlying directional means are in Table 9.

thrust model	|residual| [mN m]				magnitude deficit	
	median	p90	max	RMS	mean [mN m]	one-sided
(a) no ground effect	204.8	301.5	349.5	213.3	−199.4	20/20
(b) GE, no rotor interference	163.9	258.9	304.2	176.1	−159.1	20/20
(c) GE with rotor interference	**61.0**	**127.6**	**177.4**	**86.0**	**−35.8**	12/20

Table 11: Residuals of the identified directional means against the static thresholds (odometry-fitted arms held fixed). The magnitude deficit is $|M_{\mathrm{ident}}| - |M_{\mathrm{pred}}|$, whose sign exposes systematic over-prediction (the raw signed residual hides it, the two tip directions carrying opposite signs); one-sided" counts the groups over-predicted. Each added physics element removes part of a strictly one-sided bias, and the parameter-free interference model leaves no one-sidedness worth the name.

than any plausible systematic on where the normal-force resultant acts within a compliant contact patch. Nothing about the aerodynamic model can be resolved beneath that floor, which is why this section reports a budget and not a fit.

Applying the correction does not improve the deliverable. The two channels reach the offset unequally, and the arithmetic is exact. Pairing the tip directions in the onset balance gives $S_{\mathrm{off}} = (1 + b)\, M_{\mathrm{ff}} + (a_+ - a_-)/2 - [(W - f_+)l_+ - (W - f_-)l_-]/2$. The thrust channel is antisymmetric and cancels, leaving 0.111 mm in the median from its direction asymmetry alone; the moment-proportional channel multiplies M_{ff} and does not cancel at all, scaling the delivered offset by 4.31% — 0.413 mm in the median and 0.654 at worst, beside a 1.64 mm validation RMS. It is therefore large enough to matter, and it can be tested directly: applying it makes the agreement with the load-cell truth *worse*, the RMS rising from 1.64 to 1.82 mm and the mean error from −0.91 to −1.20. This is the degeneracy above seen from the deliverable's side. A contact-geometry term of comparable size acts the other way, the two cannot be separated at this arm precision, and correcting for one alone therefore removes a real effect while leaving its counterpart in place. The channel is accordingly reported as a bound on the deliverable and not applied to it (`analysis/ge_offset_effect.py`); the identification is left uncorrected, and the 1.64 mm is quoted as measured.

That limitation is immaterial to the method, and by construction. The fidelity analysis assumes only that the perturbation is smooth and thrust-proportional — which *both* candidates are — and shows that any such term enters as a constant, a τ-linear and a φ-linear contribution plus a bounded remainder, i.e. it renames the effective constants that the calibration estimates rather than deforming the cosh family. Both candidates also enter the two tip directions antisymmetrically through $\pm(W - f)\, l$ and $\pm a$, so both cancel in the pivot-free deliverable (6). This is why M_{ff} validates to millimetres (Sec. 9) even though the individual thresholds carry a ~ 0.2 N m ground-contact budget whose internal split this experiment cannot resolve.

			l_{odom}	COSH	no GE		no int.	with interference		benchmark
case	axis	dir	[mm]	$\bar{M}$ [N m]	pred	resid	resid	pred	resid	range [N m]
1	x	-	145.2	−1.051	−1.401	+350	+304	−1.217	+165	[−1.048, −1.018]
1	x	+	139.2	+0.950	+1.172	−222	−180	+1.002	−52	[+0.894, +0.977]
1	y	-	109.3	−0.421	−0.630	+209	+179	−0.508	+87	[−0.476, −0.373]
1	y	+	120.1	+1.090	+1.431	−341	−301	+1.268	−177	[+1.054, +1.130]
2	x	-	140.0	−1.829	−1.933	+104	+54	−1.731	−98	[−1.861, −1.680]
2	x	+	139.4	+0.870	+1.019	−149	−109	+0.856	+14	[+0.789, +0.913]
2	y	-	105.0	−0.675	−0.791	+116	+86	−0.667	−8	[−0.714, −0.641]
2	y	+	118.5	+1.338	+1.566	−228	−187	+1.398	−60	[+1.248, +1.333]
3	x	-	142.5	−1.542	−1.672	+130	+82	−1.479	−63	[−1.608, −1.339]
3	x	+	138.3	+0.997	+1.294	−297	−254	+1.121	−123	[+1.033, +1.075]
3	y	-	109.3	−0.984	−1.253	+270	+233	−1.106	+123	[−1.059, −0.990]
3	y	+	117.9	+0.937	+1.145	−208	−171	+0.995	−58	[+0.923, +1.014]
4	x	-	141.1	−1.136	−1.280	+144	+101	−1.104	−32	[−1.181, −1.151]
4	x	+	138.4	+1.468	+1.674	−206	−159	+1.484	−16	[+1.406, +1.516]
4	y	-	106.7	−0.998	−1.202	+204	+169	−1.060	+62	[−1.044, −0.928]
4	y	+	118.3	+0.982	+1.174	−192	−154	+1.022	−40	[+0.969, +1.058]
5	x	-	142.5	−1.019	−1.158	+140	+97	−0.987	−32	[−1.048, −0.932]
5	x	+	137.9	+1.695	+1.804	−109	−61	+1.609	+86	[+1.624, +1.695]
5	y	-	103.2	−0.645	−0.741	+96	+66	−0.620	−25	[−0.667, −0.539]
5	y	+	118.9	+1.323	+1.601	−278	−236	+1.432	−109	[+1.241, +1.351]

Table 12: Per-direction static-threshold check with odometry-fitted arms; residuals in mN m. The last column pair is the parameter-free rotor-interference model; the rightmost entry of each row is the range spanned by the four alternative estimators (NLS, CPD_N, CPD_R, CUSUM), showing that the disagreement with the static prediction is not an artifact of the onset detector.

11 Reproducibility

Every number in this document is reproducible from the repository:

- `critical_value_getter_piecewise.py` — pipeline, gates, caps, calibration, Algorithm 1;
- `analysis/ramp_quality.py` — gates;
- `analysis/sensitivity.py` — ±20% robustness;
- `analysis/cosh_fidelity.py` — prediction-test residuals;
- `analysis/excitation_angle_design.py` — rate/cutoff design;
- `analysis/ge_linearity.py`, `analysis/ge_trajectory.py` — ground-effect bracket;
- `analysis/tiltcap_ablation.py` — window ablation;
- `analysis/baseline_stat_ablation.py` — median/mean/zero baseline;
- `analysis/bias_drift.py` — in-window baseline drift;
- `analysis/nls_comparison.py` — Tables 7–10;
- `analysis/mcrit_prediction.py`, `analysis/mcrit_static_figure.py`, `analysis/static_attribution.py` — Sec. 10 (static-threshold budget, arms, attribution limits, Fig. 5);
- `analysis/ge_bilinear.py` — the bilinear $\tau\varphi$ ground-effect term along measured trajectories;
- `analysis/constrained_calibration.py` — Remark 2 and Table 5 (the shared $(W z_{\text{CoM}}, J_{P,\text{roll}}, J_{P,\text{pitch}})$ reparametrization and the rerun identification);
- `analysis/cv_table.py` — Table 3 (rate consistency under the free, CAD and mis-specified constants);
- `analysis/phys_compare.py` — Table 4 (calibrated constants against the CAD reference and the parallel-axis floor);

- `analysis/cad_reference.py` — Table 6 (the identification rerun with nothing fitted);
- `analysis/pivot_geom.py` — the mocap arc fit for l_p and the marker height;
- `analysis/pnls_centred.py`, `analysis/pnls_ci.py` — the two-stage calibration (per-run PNLS medians, then the score refined in a neighbourhood of them) and its intervals;
- `analysis/mff_table.py`, `analysis/cv_full.py` — the estimator benchmark extended with the PNLS-centred and CAD constants, and the full-pipeline dispersions;

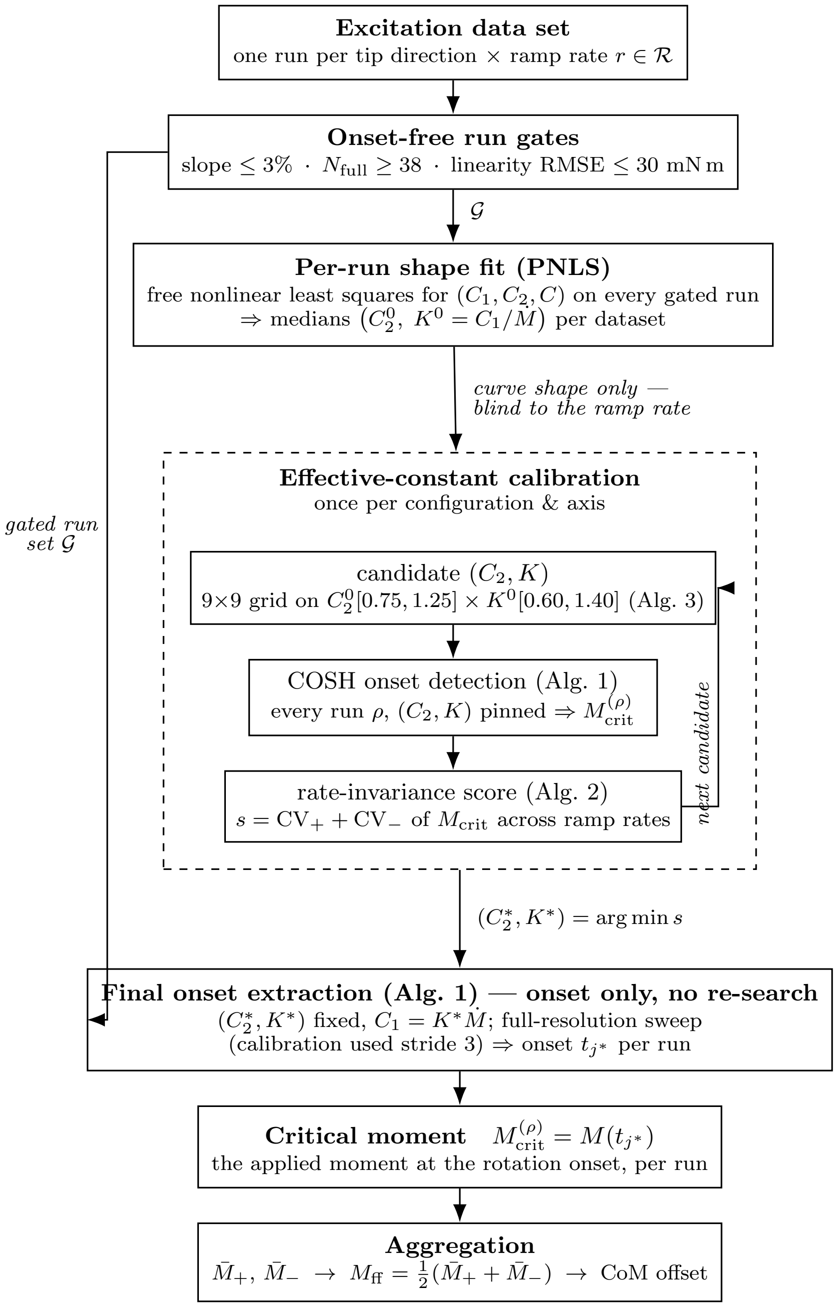

Figure 1: Pipeline at a glance. The dashed container is the once-per-configuration calibration: it iterates candidate pairs (C_2, K), running the *same* onset detector (Algorithm 1) on every gated run and scoring the candidates by rate-invariance of the resulting critical moments. The selected pair then drives one final onset extraction per run — the constants are *not* re-estimated; only the onset index is re-swept, at full resolution, because the calibration subsamples the sweep by a stride of 3 for speed — whose onset sample yields $M_{\text{crit}} = M(t_{j^*})$, and the directional means aggregate into the pivot-free feed-forward moment.

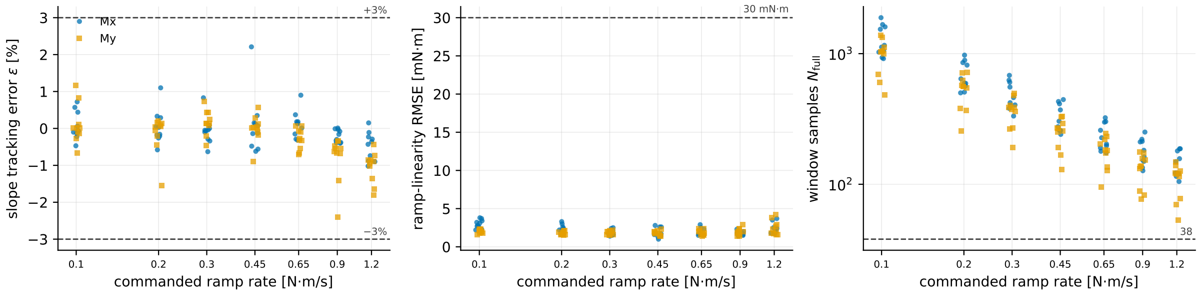

Figure 2: Run-gate quantities for all 140 runs versus the commanded ramp rate, in gate order: slope tracking error ε and ramp-linearity RMSE on the floored segment ($|M| \geq M_{\text{floor}}$), and the excitation-window sample count N_{full}. Dashed lines are the gate thresholds (±3%, 30 mN m, 38 samples); every run clears all three.

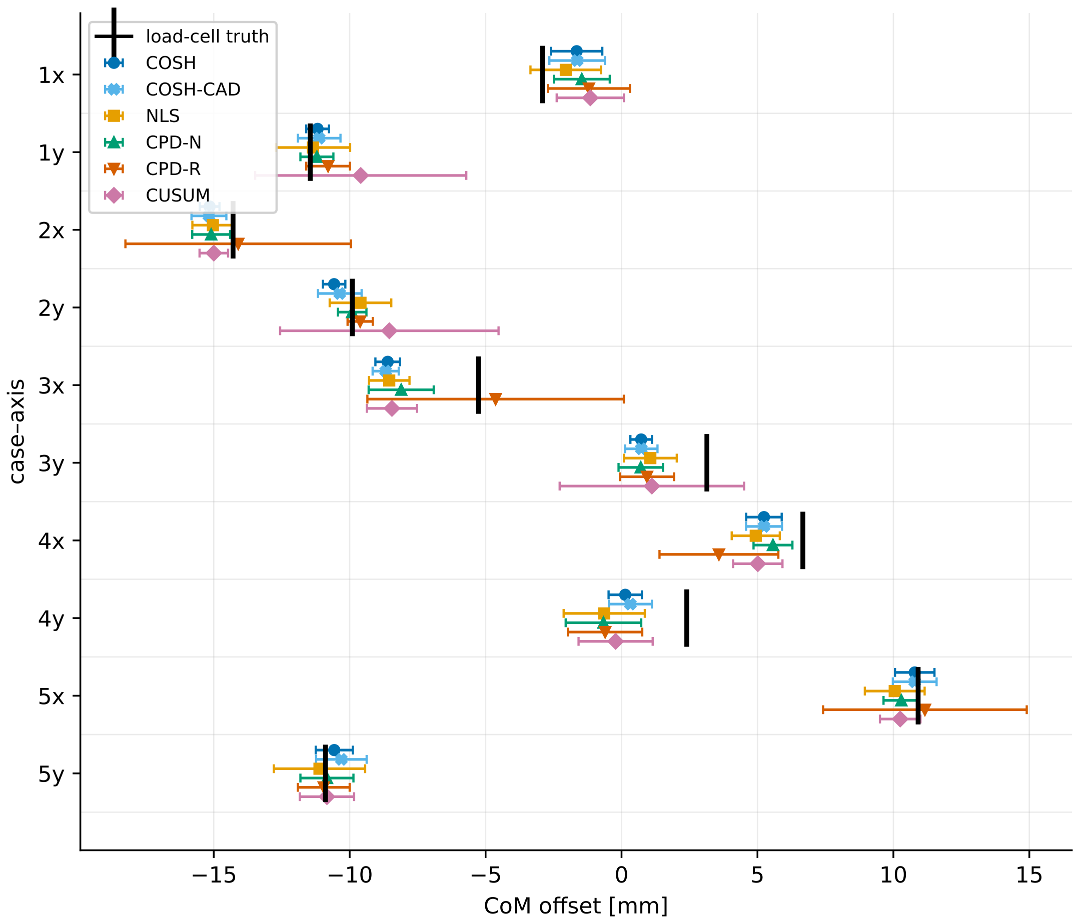

Figure 3: Estimated CoM offset with 95% confidence intervals for six estimators, against the load-cell truth (black tick), per case–axis pair. COSH (dark blue) has the narrowest interval in 9 of 10 pairs (±0.36–0.94 mm); COSH_{CAD} (light blue) tracks it closely in the offset while carrying a wider interval, which is the rate-consistency penalty of Table 10. The nonparametric detectors blow up to ±4–5 mm on the pairs where they misplace onsets (CPD_{R} at 2x, 3x, 5x; CUSUM at 1y, 2y, 3y). $\text{COSH}_{\text{wide}}$ is omitted here because it is visually indistinguishable from COSH; it appears in the tables. Marker shapes duplicate the color coding.

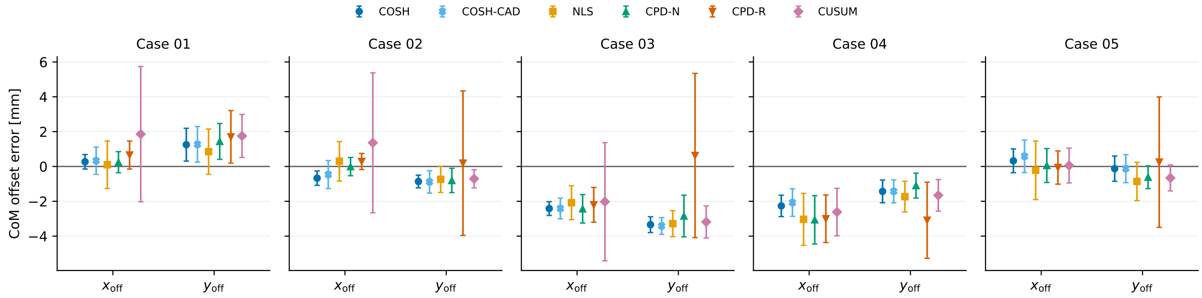

Figure 4: The same data folded about the truth: CoM offset *error* (estimate − truth) with 95% confidence intervals, grouped by component within each case (x_{off} from the pitch excitation, y_{off} from the roll excitation). The residual biases of cases 3 and 4 are *common to every estimator* — a vehicle–contact asymmetry, not a detection artifact — while the estimator-to-estimator differences stay within about ±1 mm.

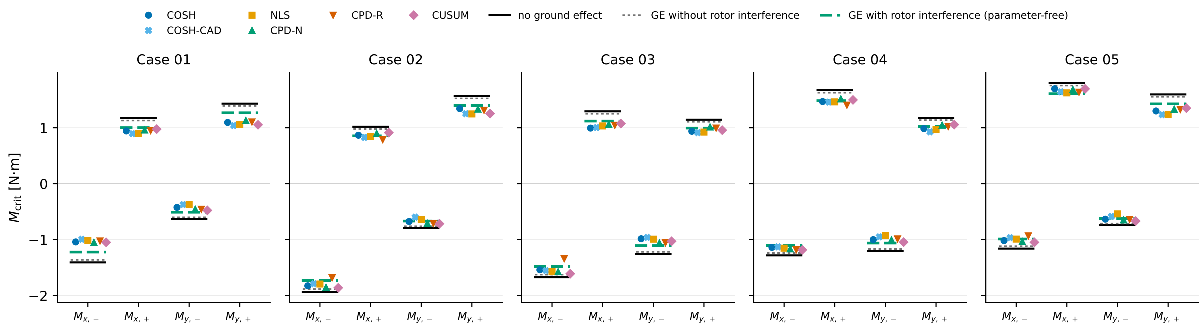

Figure 5: Identified directional critical moments (five estimators, colored markers) against the static-threshold predictions with truth offsets and odometry-fitted arms: no ground effect (black solid), ground effect without rotor interference (gray dotted), and with rotor interference (green dashed). The interference prediction falls inside the range spanned by the five estimators in 9 of the 20 groups, where neither of the other two ever does (0/20); the median estimator range is 108 mN m. A 19 mm inboard shift of the static contact lever would move the black line by a similar amount: the figure shows the *size* of the near-ground correction, not its attribution.